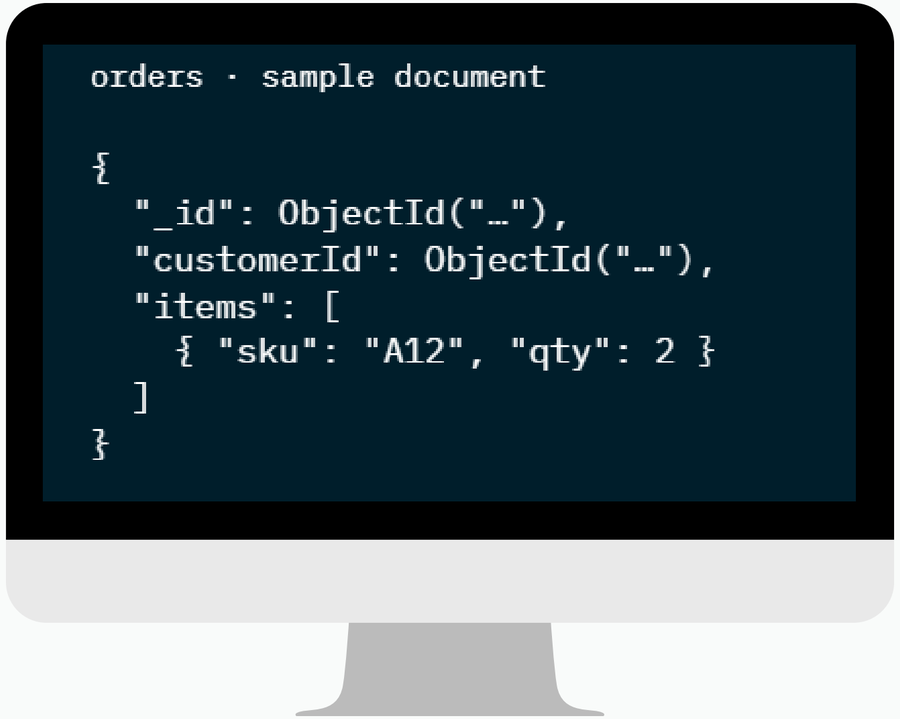
Applied Data Science Project • Checkpoint 1

Dear LLM, Can You Explain my MongoDB Database?

Project design: problem, users, requirements and plan



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Mentors: Silvia Chiusano, Alessandro Fiori

Project Goal



THE PROBLEM

MongoDB databases rarely explain themselves, for this reason it is challenging to:

- Understand main database design choices
- Migrate and review databases safely
- Exhaustively document databases

THE IDEA

Use LLMs as a reliable starting point for users, providing evidences and confidence levels of the output.

Value Proposition

Decent Work and Economic Growth
Experts can exploit LLMs as a starting point to interpret complex databases.

Industry, Innovation and Infrastructure
Documenting migrations improves reliability and durability of the digital infrastructure.

Quality Education
Helps students to learn how to model databases and understand the main design choices.



Main Tasks



PATTERN RECOGNITION

Identify the pattern among: *Embedding, Referencing, Subset, Arrays or Bucket.*

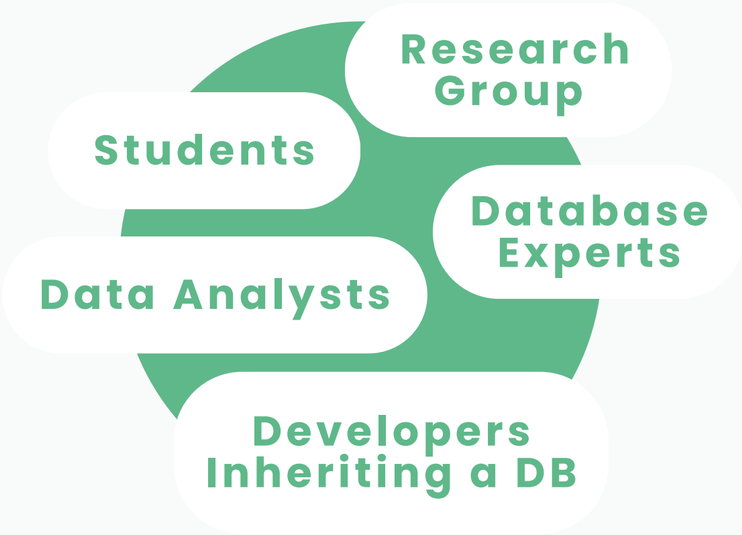
EVIDENCE AND REASONING

Provide evidences and explanations behind every design choice.

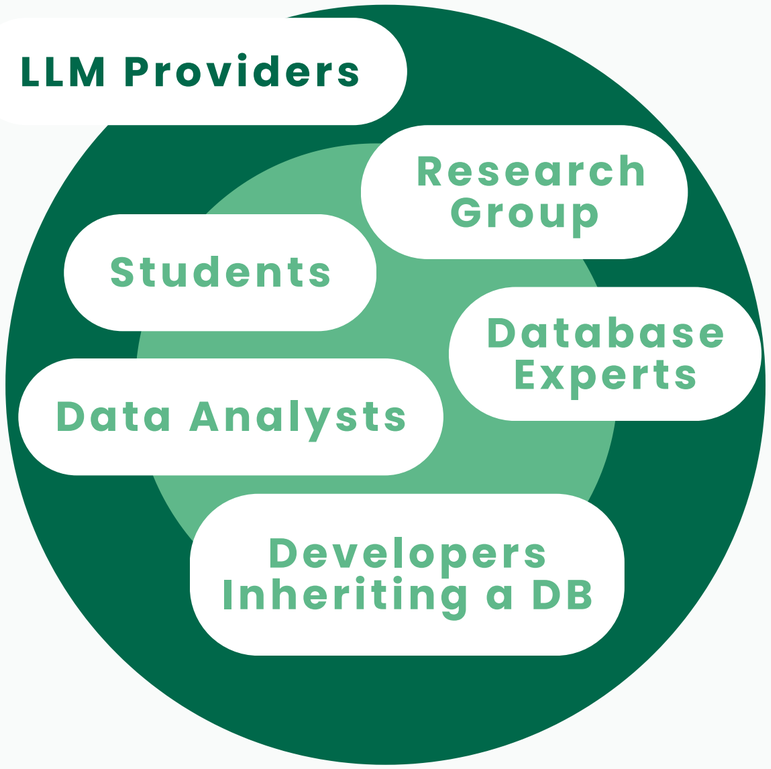
EXPECTED OUTPUT

The detected pattern with a confidence level: *Confirmed, Plausible, Undeterminable*

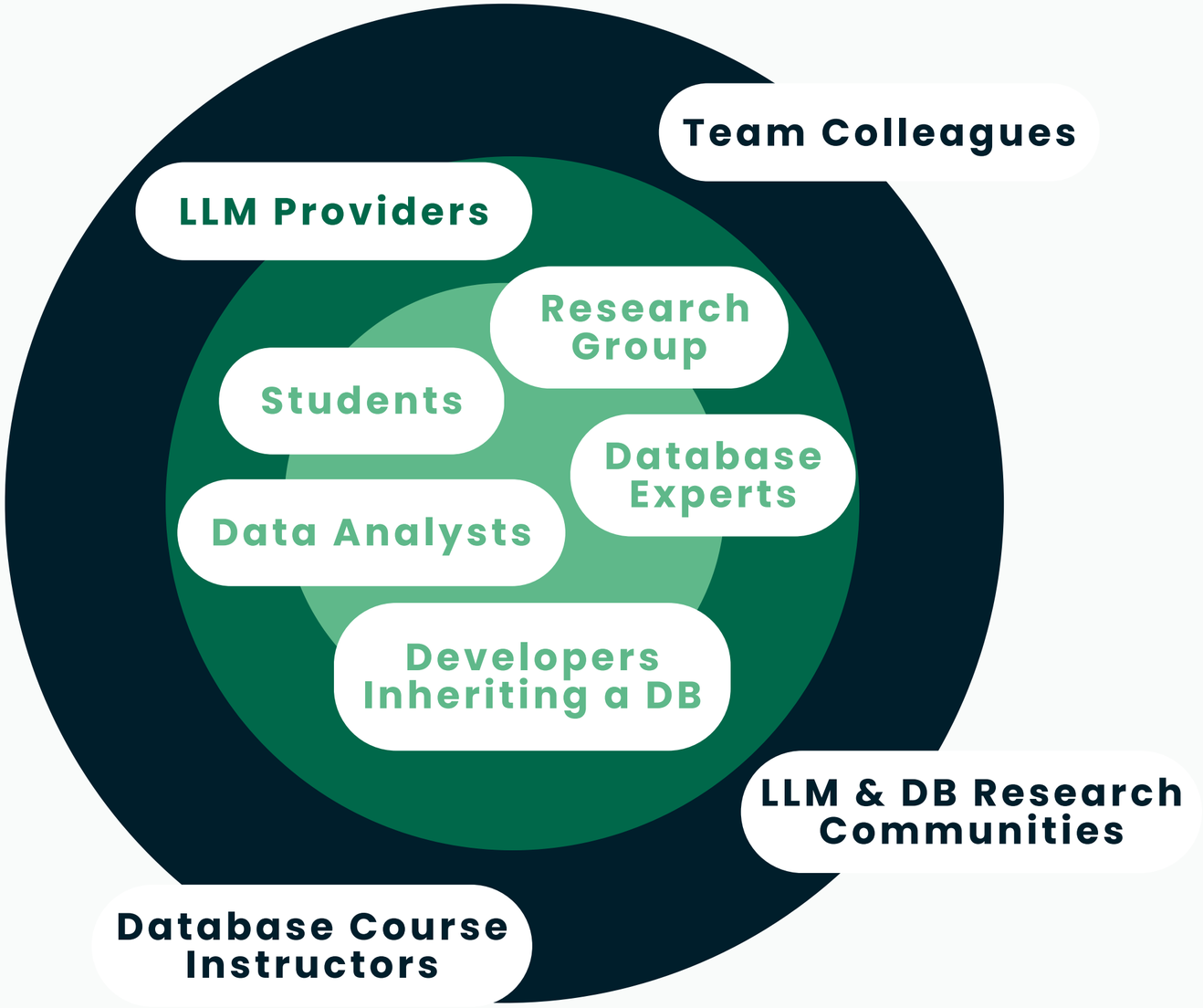
Stakeholder Map: Direct Users



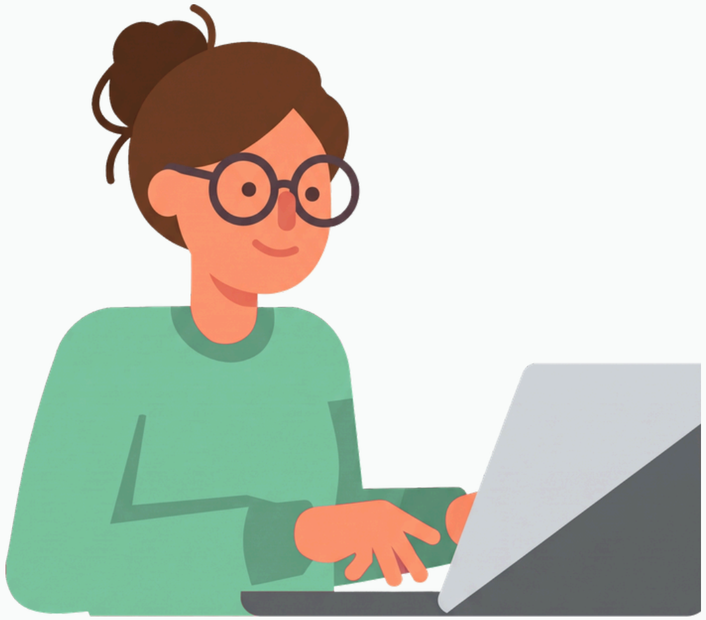
Stakeholder Map: External Influencers



Stakeholder Map: Indirect Beneficiaries



User Personas



STUDENT IN THE DATABASE COURSE



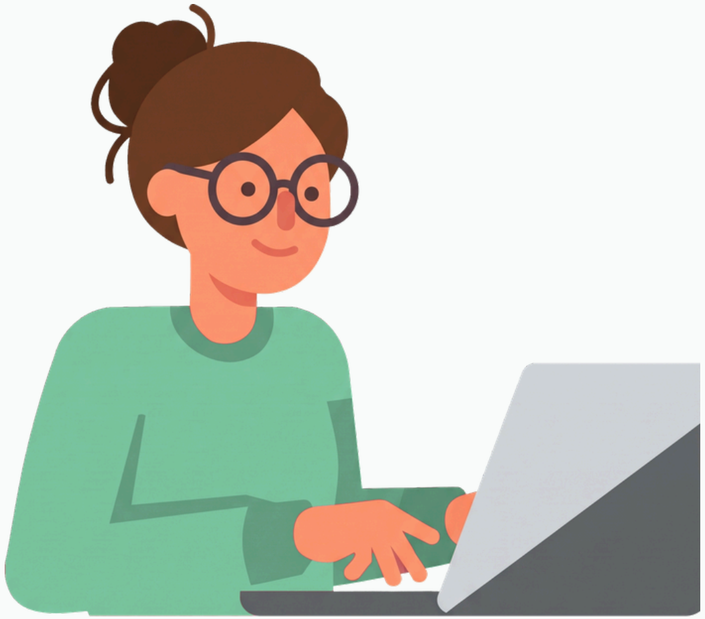
BACKEND DEVELOPER WITH 2 YEARS EXPERIENCE



DATA ARCHITECT WITH EXPERIENCE WITH MONGODB

User Personas

SARA



Goals & Needs: Understand NoSQL modeling patterns and learn to recognise them in a realistic database.

Frustrations: Lecture examples are clean, real databases are not; confuses subset and embedding, or any array with a pattern.

How She Works Today: Rereads the slides, searches online, asks classmates or a generic chatbot. The risk is too much trust on the AI.

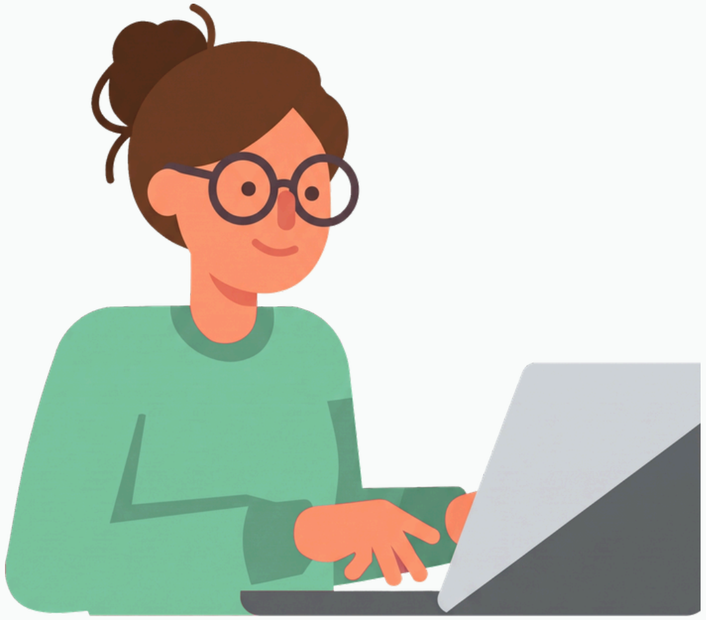
User Stories: Sara

MUST

“As a student, I want to see next to each pattern its definition and the field that proves it, so that I can learn to recognise it on my own.”

MUST

“As a student, I want the system to say clearly when it is not sure, so that I don’t learn something wrong”



User Personas

LUCA

Goals & Needs: To quickly understand the DB he has to modify.

Frustrations: Implicit references (scattered ObjectIds), obscure nested arrays, ambiguous field names.

How He Works Today: Opens random collections, reads sample documents, searches field names in the code, asks colleagues. The risk it to give too much trust to collegue’s assumptions.



User Stories: Luca

MUST

“As a developer inheriting a DB, I want a map of collections, fields and relationships, so that I can find my way around quickly.”

MUST

“As a developer, I want every relationship and pattern to point to the fields it comes from, so that I know what to check before changing the code.”



User Personas

ELENA

Goals & Needs: Evaluate the data model prior to an update or migration, and document it.

Frustrations: Slow manual analysis, Lacks trust in answers that are confident yet incorrect

How She Works Today: Starts from key collections, maps out relationships, and analyzes access patterns.



User Stories: Elena

MUST

“As a data architect, I want to see evidence and alternatives for every attribution, so that I can verify it before a migration.”

MUST

“As a data architect, I want uncertain attributions to be clearly highlighted, so that I know which ones to check myself.”



User Requirements

Functional Requirements

INPUT:

JSON schema + sample documents

STRUCTURE:

Collections, fields, relationships

PATTERNS:

The 5 modelling patterns

EVIDENCE:

verified fields + alternatives

CONFIDENCE:

Confirmed, Plausible, Undeterminable

DOCUMENTATION:

Comprehensive schema explanation

User Requirements

Non Functional Requirements

REPRODUCIBILITY:

Prompt, models, parameters logged

RESOURCES:

Open-source model

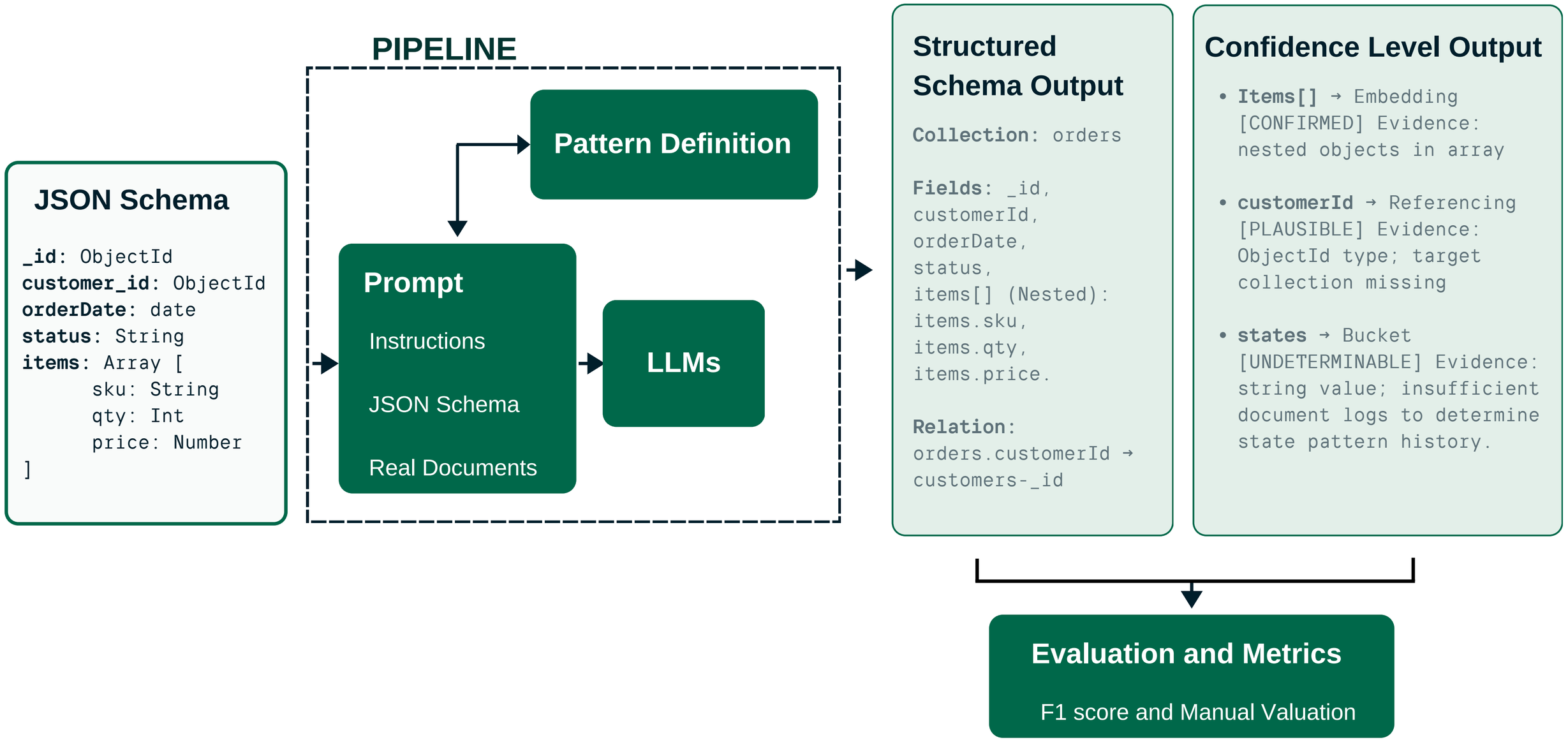
PRIVACY:

No personal data sent via API

TRANSPARENCY:

Model and inputs declared

Functional Diagram



Datasets



STRATEGY: Synthetic data generation with expert validation

- **Varied domains**: e-commerce, IoT, hospital, streaming.
- Mixed, **increasing difficulty**; all 5 patterns covered.
- Edge cases: missing or ambiguous data to test all three confidence levels.

Evaluation and Metrics



METRICS

- **Structure and Pattern F1:** Automatic verification of identified collections, fields, relationships, and the 5 modeling patterns (plus their confidence level)

EVALUATION

- **Confidence & Reasoning Review:** Manual check and evaluation of LLM's explanation and confidence levels with respect to data evidence

- **Cross-Model Comparison:** Performance differences and consistency across the 2–3 tested LLMs.

Risks



CONTEXT WINDOW SATURATION

Large, nested sample documents can exceed the context of open-source models.



HALLUCINATIONS AND OVERCONFIDENCE

Convincing explanations that cite fields which do not exist.



PATTERN CONFUSION

An embedded address object read as a reference, or the reverse.

Timeline Roadmap



CHECKPOINT 1 (CURRENT STATUS)

Requirements & problem definition; system design.

PHASE 1 • BENCHMARK AND SETUP

Synthetic data generation & ground truth validation.

PHASE 2 • PIPELINE & CORE EVALUATION

Prompting pipeline implementation & automated metrics.

PHASE 3 • COMPARISON & FINAL

Cross-model benchmark & manual reasoning review.

THANK YOU
for your attention